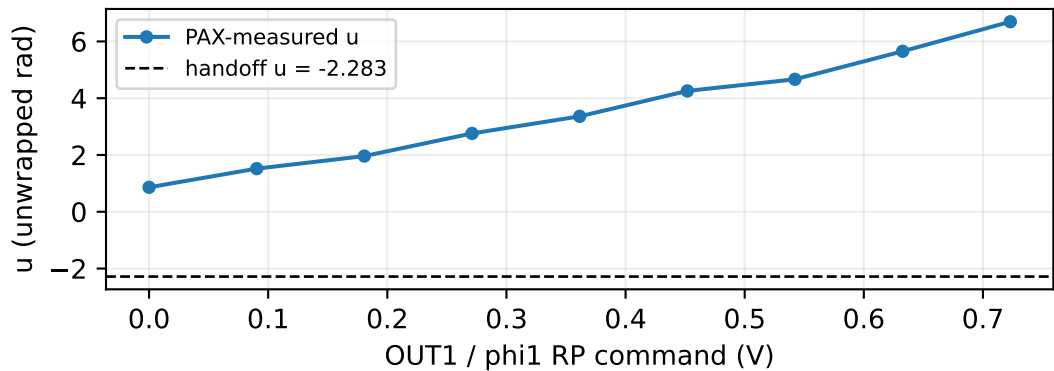
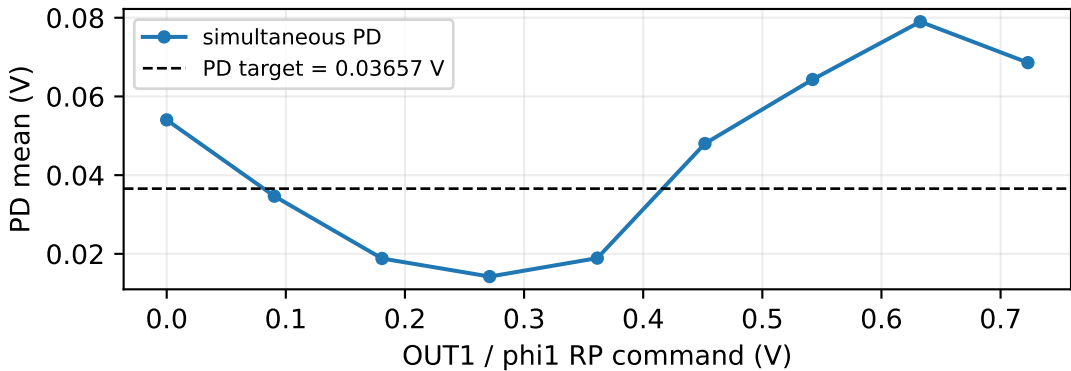


Phi1 PD FPGA lock — 021210_phi1-pd-hybrid-test_autonomous-hybrid-gentle-1

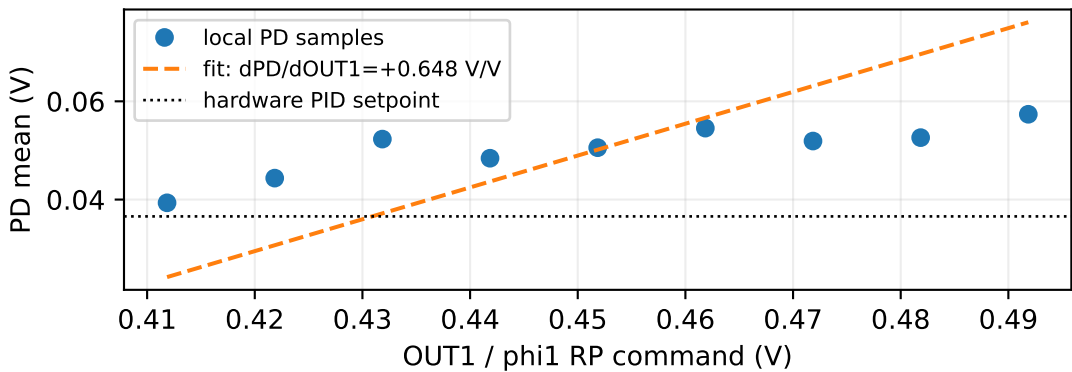
PAX-assisted $\phi 1$ rough map



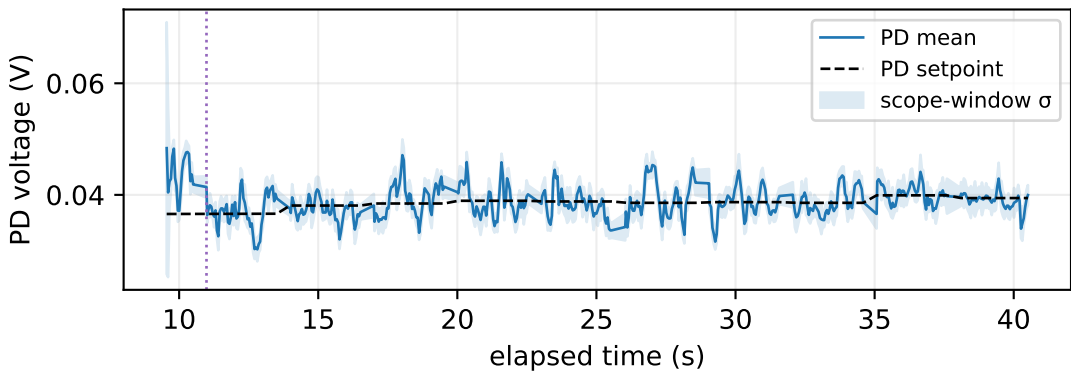
PD value associated with rough u map



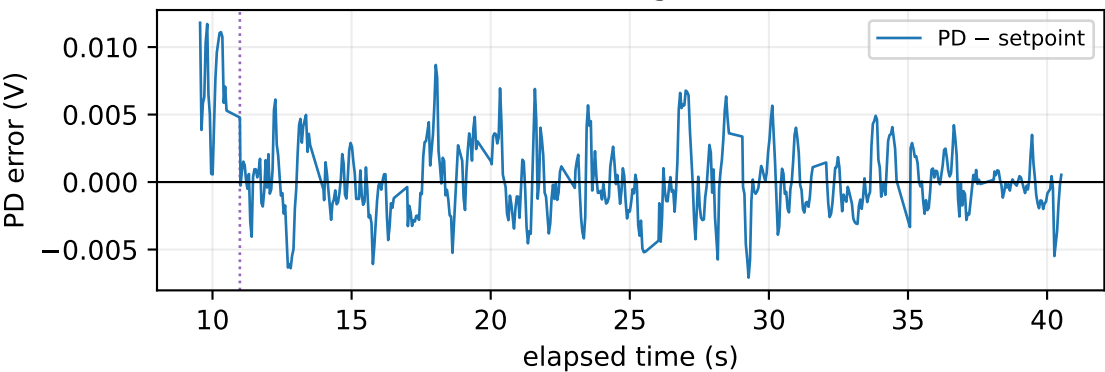
Automatic local PD calibration



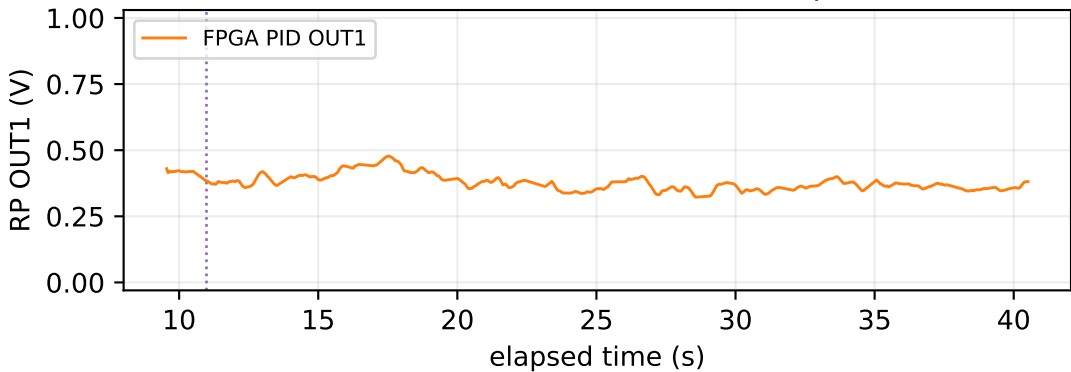
FPGA PD lock



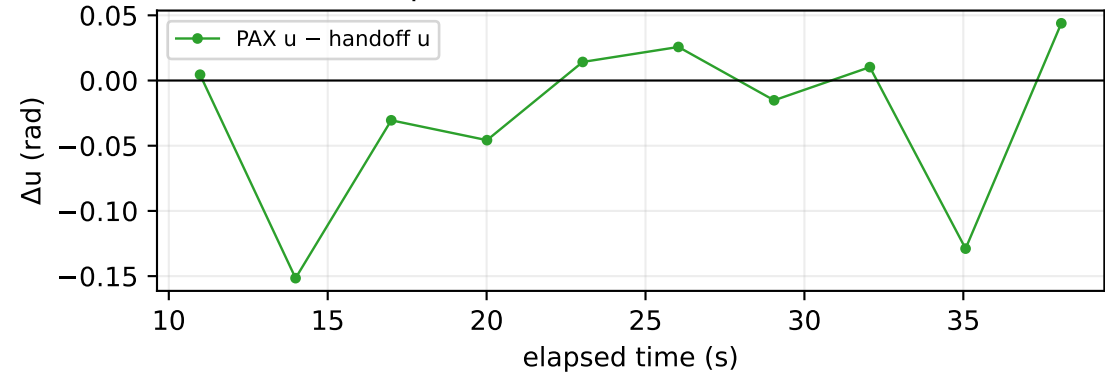
PD locking error



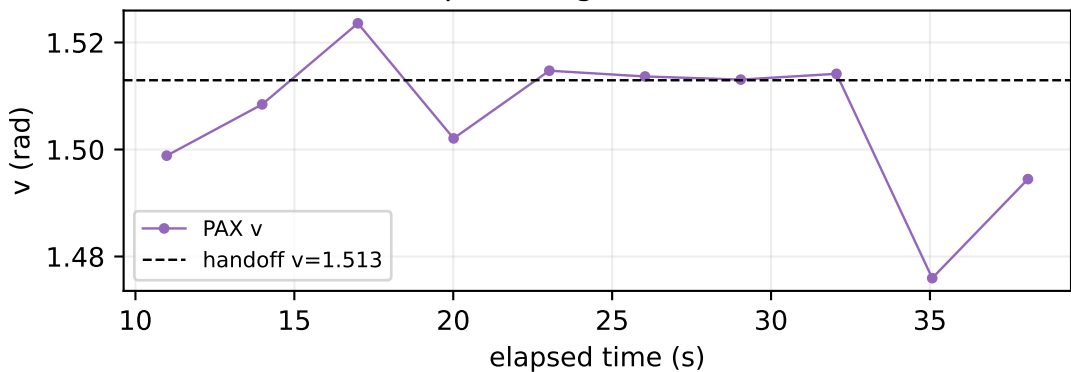
Continuous FPGA actuator output



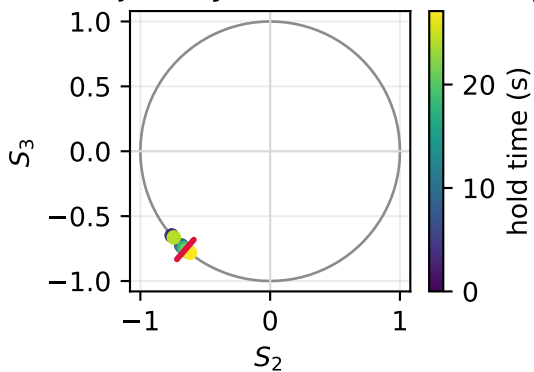
Independent PAX azimuth validation



PAX polar-angle cross-check



Held lock trajectory on the Poincare equator



PAX handoff target $u = -2.2828$ rad | initial PD setpoint = 0.03657 V | local slope = +0.6485 V/V
FPGA PID: $P = -0.339$, $I = -5$ Hz | duration = 31.0 s | median |PD error| = 0.00157 V (late 0.00136 V)
OUT1 = 0.322...0.477 V | PD setpoint = 0.03657...0.03992 V | PAX median $|\Delta u| = 0.028$ rad (10 snapshots)